Leaf Mass on the Sunny and Shady Sides of a Tree

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2026-11-10

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is a comment in the quarto file 1. Save this file at the TOP of your project folder (next to data/, themes/, figures/, scripts/) and name it leaf_report.qmd 2. Press Preview in the upper left of the editor window right away — it should build before you add anything. 3. The analysis code is already here and runs. Your job: add the finishing touches marked # TODO: (colors, labels, theme, table formatting), write the text sections, and render as you go.

KEEP THIS FILE. For any future analysis, copy it, change the data file in the load-data chunk and the column names, and rewrite the text.

Paths like "data/…" work because Quarto runs the code from the folder this file is saved in — the top of your project. —>

1 Introduction

- **Null hypothesis:**
- **Alternate hypothesis:**

2 Methods

3 Results

3.1 Box plot

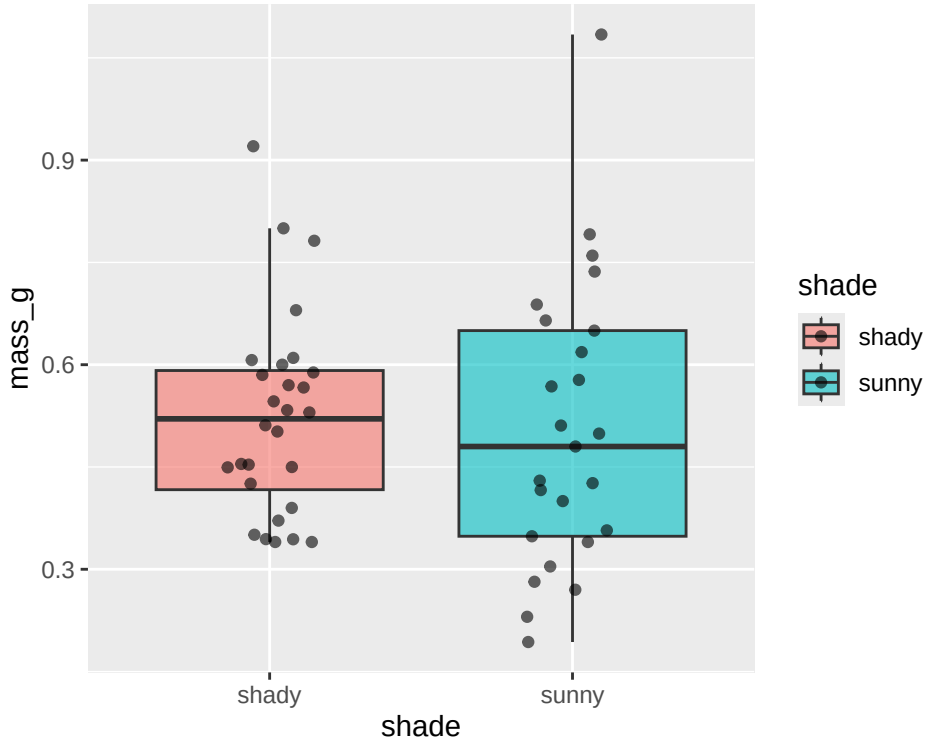

Figure 1: Leaf mass by side; each point is one leaf.

3.2 Mean ± SE

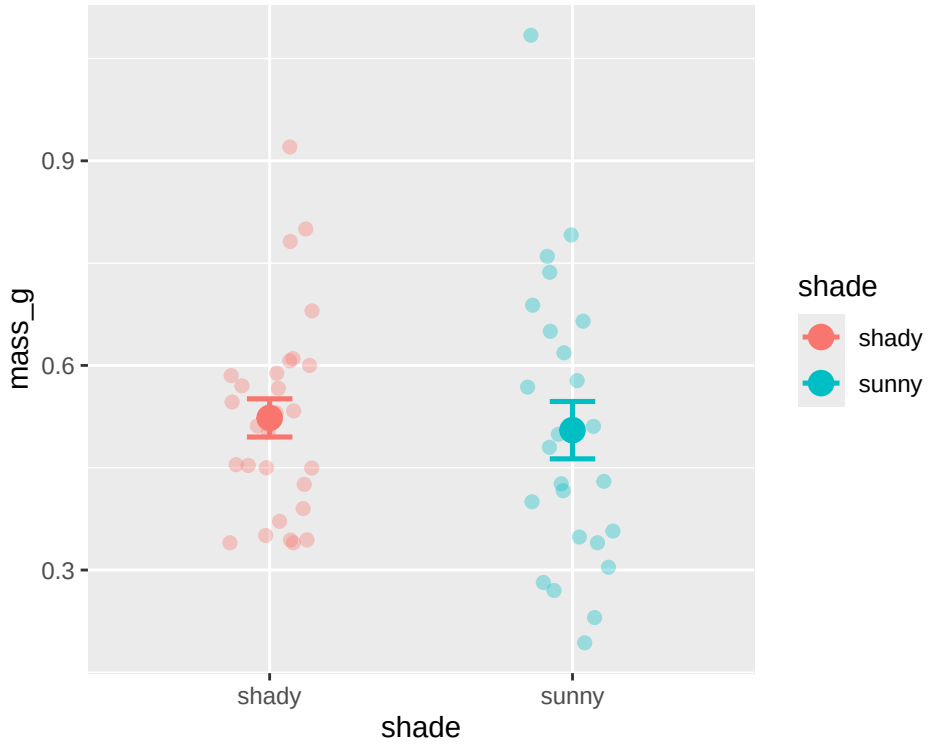

Figure 2: Mean leaf mass (± 1 SE) by side.

3.3 Histogram

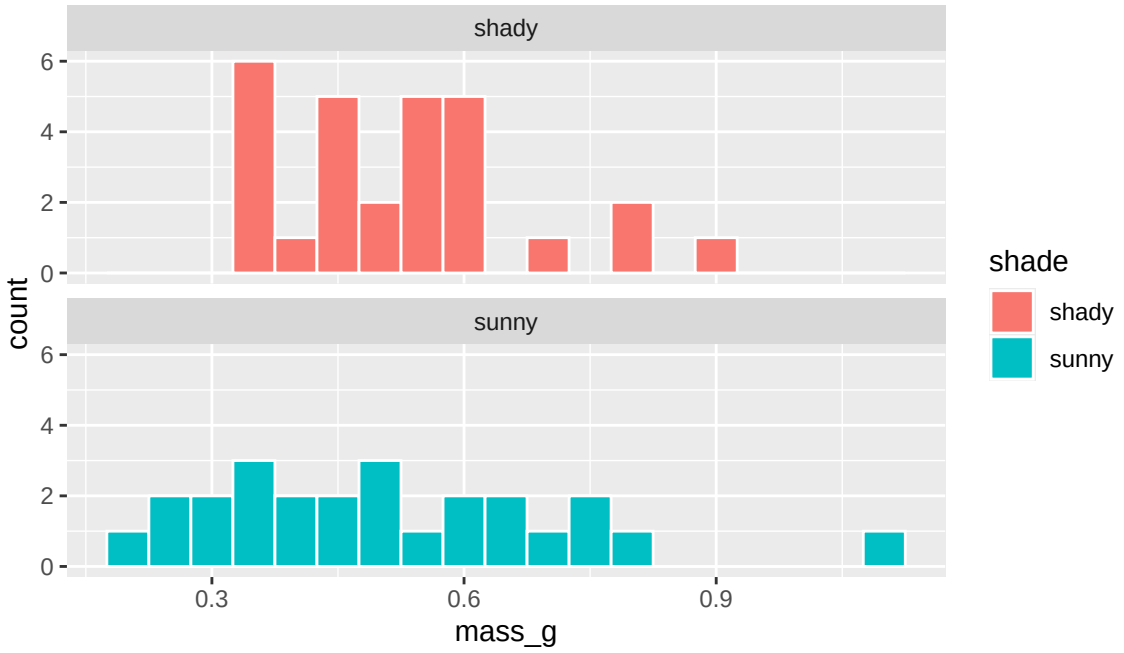

Figure 3: Distribution of leaf mass on each side.

3.4 Summary statistics

Table 1: Summary statistics for leaf mass (g) by side.

shade	n	mean	variance	sd	se	ci_lower	ci_upper
shady	28	0.5230357	0.0219048	0.1480026	0.0279699	0.4656463	0.5804251
sunny	25	0.5050640	0.0442281	0.2103048	0.0420610	0.4182544	0.5918736

3.5 Welch's t-test

```
    Welch Two Sample t-test

data:  mass_g by shade
t = 0.35579, df = 42.527, p-value = 0.7238
alternative hypothesis: true difference in means between group shady and group sunny is not e
```

```
95 percent confidence interval:
 -0.08392774  0.11987117
sample estimates:
mean in group shady mean in group sunny
          0.5230357           0.5050640
```

Table 2: Welch's t-test of leaf mass by side of the tree.

estimate1	estimate2	statistic	parameter	p.value	conf.low
0.5230357	0.505064	0.3557927	42.52693	0.7237547	-0.0839277

4 Discussion